

Muthulakshmi M

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Education

Rajalakshmi Engineering College, Tamil Nadu, India

B.Tech in Artificial Intelligence and Data Science

CGPA: 8.09

Sep 2023 – June 2027

Current Semester: 6

Skills Summary

Programming Languages: Python, SQL, Java (Basic), C, JavaScript, HTML, CSS, R

AI & Machine Learning: TensorFlow, Scikit-Learn, CNN, NLP, Deep Learning, Grad-CAM

Data Analytics & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Excel

Web & Backend Development: React.js, Node.js, Express.js, Flask, REST APIs

Databases & Cloud Platforms: MySQL, PostgreSQL, MongoDB, SQLite, Snowflake

Tools & Platforms: GitHub, Jupyter Notebook, VS Code, IntelliJ IDEA, Android Studio, Figma, Render, Vercel

Soft Skills: Leadership, Communication, Problem Solving, Team Collaboration

Work Experience

Artificial Intelligence Intern

- Applied AI techniques for model development and problem-solving tasks.

Full Stack Development Intern

- Built and optimized web applications using front-end and back-end technologies.

UI/UX Design Intern

- Designed user-friendly interfaces using wireframes and prototypes.

Projects

AI-Based Currency Detection System for Visually Impaired People

Jan 2026 – Apr 2026

- Built a deep learning-based Indian currency recognition system using TensorFlow and MobileNetV2.
- Developed a Flask web application supporting camera capture, image upload, and real-time prediction.
- Implemented voice-assisted accessibility features for visually impaired users.
- Improved model robustness using transfer learning and data augmentation.

Global Energy & Nuclear Safety Analytics System — Team Project

Feb 2026 – Apr 2026

- Performed data cleaning, preprocessing, and exploratory data analysis on global energy datasets.
- Analyzed power plant, carbon emission, fuel distribution, and radiation monitoring datasets.
- Created visualizations and trend analysis using Pandas, Matplotlib, and Seaborn.
- Developed interactive dashboards and reports using Tableau.

S.C.O.P.E.S – AI Powered Smart Price Comparison & Recommendation System

Jan 2026 – Apr 2026

- Developed a full-stack AI-powered price comparison platform using React.js, Node.js, MongoDB, and Flask.
- Implemented REST APIs, TF-IDF product matching, and recommendation systems.
- Integrated machine learning models for AI-based price prediction and deal score analysis.
- Designed cache-first architecture for optimized real-time comparison.

MGM Resort Security & Insider Threat Detection System — Team Project

Jan 2026 – Mar 2026

- Developed a secure cyber security platform using Python, Flask, and MongoDB.
- Implemented role-based authentication, session management, and password hashing.
- Built insider threat detection and honeypot attack simulation modules.
- Developed real-time security dashboards and alert systems.

Lung Cancer Detection using CNN and Grad-CAM — Team Project

Sep 2025 – Nov 2025

- Developed a deep learning-based lung cancer detection system using CNN.
- Performed preprocessing, segmentation, and data augmentation on CT scan datasets.
- Implemented Grad-CAM visualization for explainable AI and tumor highlighting.
- Achieved 94.6% classification accuracy using TensorFlow and Keras.

AstroGeo AI – Intelligent Chatbot for Earth Geography & Space Exploration

Feb 2026 – Apr 2026

- Developed an AI-powered chatbot using NLP techniques for geography and space exploration queries.
- Integrated datasets including countries, rivers, planets, earthquakes, and climate data.
- Built semantic search and knowledge retrieval using Sentence Transformers and FAISS.
- Developed a conversational web application using React, Flask, and Tailwind CSS.

Certifications

• SnowPro Associate: Platform – Snowflake	Apr 2026
• Introduction to Tableau – Simplilearn	Mar 2026
• Azure DevOps for Beginners – LinkedIn Learning	May 2025
• Microservices: Design Patterns – LinkedIn Learning	May 2025
• DevOps Foundations – LinkedIn Learning	May 2025
• Machine Learning – LinkedIn Learning	Nov 2024
• R for Data Science – LinkedIn Learning	Apr 2025
• Programming Using Java – Infosys Springboard	Apr 2024
• The Complete Guide to HTML & CSS – Udemy	Mar 2025